

MENGJUN WANG

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RESEARCH INTERESTS

Task-grounded Physical AI for construction and infrastructure; construction automation and robotics; task-relevant sensing and physical-state estimation; digital-to-physical interfaces and spatial guidance; adaptive autonomy; human-AI-robot collaboration.

EDUCATION

- **University of Florida** 01/2025 – Present
Ph.D. in Civil Engineering, Expected May 2027
◦ GPA: 4.00/4.00 | Advisor: Dr. Shuai Li
◦ Dissertation Proposal: *Physical AI for Industrialized Construction: From BIM Design to Workface Guidance and Robot Action*
◦ Began doctoral studies at the University of Tennessee (08/2021–12/2024) and transferred to the University of Florida with advisor in January 2025.
- **University of Tennessee** 2024
M.S. in Computer Science; Minor in Statistics Knoxville, TN
◦ GPA: 3.89/4.00
- **University of Tennessee** 2024
M.S. in Civil Engineering, Thesis-based Knoxville, TN
◦ GPA: 4.00/4.00
- **Changsha University of Science and Technology** 2016 – 2020
B.S. in Traffic Engineering Changsha, China
◦ GPA: 3.68/4.00 | Rank: 1/73

PUBLICATIONS

Peer-Reviewed Journal Articles

- [8] **Wang, Mengjun**, Li, Shuai, and Zhang, Yunlan. (2026). Robotic localization via active projected fiducials for space construction. *Journal of Computing in Civil Engineering*. In Production.
- [7] Uthai, Thanakon, You, Hengxu, **Wang, Mengjun**, Smith, Kaleb, Spackman, Everett, Ryan, Zoe, Li, Shuai, and Du, Jing. (2026). Opportunities, challenges and roadmap for humanoid robots in construction. *Scientific Reports*, 16, 905.
- [6] **Wang, Mengjun**, Xu, Jianjun, Lassiter, H. Andrew, and Li, Shuai. (2025). BIM-driven laser spatial augmented reality for in-situ layout and assembly. *Automation in Construction*, 178, 106405.
- [5] **Wang, Mengjun**, Cai, Jiannan, Hu, Da, Hu, Yuqing, Han, Zhu, and Li, Shuai. (2025). AI-based robots in industrialized building manufacturing. *Frontiers of Engineering Management*, 12(1), 59–85.
- [4] Hu, Da, **Wang, Mengjun**, and Li, Shuai. (2025). Multi-classifier information fusion for human activity recognition in healthcare facilities. *Frontiers of Engineering Management*, 12(1), 99–116.
- [3] Li, Yan, Liu, Songyang, **Wang, Mengjun**, Li, Shuai, and Tan, Jindong. (2024). Teleoperation-driven and keyframe-based generalizable imitation learning for construction robots. *Journal of Computing in Civil Engineering*, 38(6), 04024031. **2024 Best Paper Award.**
- [2] **Wang, Mengjun**, Hu, Da, Chen, Junjie, and Li, Shuai. (2023). Underground infrastructure detection and localization using deep learning enabled radargram inversion and vision-based mapping. *Automation in Construction*, 154, 105004.
- [1] Hu, Da, Li, Shuai, and **Wang, Mengjun**. (2023). Object detection in hospital facilities: A comprehensive dataset and performance evaluation. *Engineering Applications of Artificial Intelligence*, 123, 106223.

Manuscripts Under Review and Revision

- [2] **Wang, Mengjun**, Du, Jing, Cai, Jiannan, and Li, Shuai. Robotic Alignment of Construction Components via a BIM-to-Workface Projection Interface. *Automation in Construction*. **Major revision.**
- [1] **Wang, Mengjun**, Huang, Dingyuan, Li, Shuai, Cai, Jiannan, and Hu, Yuqing. Bringing BIM-Grounded Guidance to the Jobsite: A Mobile SAR Co-Pilot for MEP Layout. *Automation in Construction*. **Under review.**

Peer-Reviewed Conference Proceedings

- [7] **Wang, Mengjun**, Mei, Ruiren, Xu, Fan, and Li, Shuai. (2026). LaserCoPilot: A Transferable Multimodal Co-pilot for Closed-Loop Installation and Assembly Guidance via Step-Indexed Laser Projection. In *Proceedings of the 43rd International Symposium on Automation and Robotics in Construction (ISARC 2026)*, 137–144.
- [6] **Wang, Mengjun**, Zhang, Xiande, Li, Shuai, Cai, Jiannan, and Hu, Yuqing. (2024). An Audio-Based Emotion Monitoring System for Enhancing Construction Worker Safety and Mental Health. In *Computing in Civil Engineering 2024*, 1000–1008.
- [5] **Wang, Mengjun**, Li, Yan, and Li, Shuai. (2024). Robotic assembly of interlocking blocks for construction based on large language models. In *Construction Research Congress 2024*, 777–786.
- [4] Hu, Da, **Wang, Mengjun**, Guo, Ruichen, and Li, Shuai. (2024). Bridge deck condition assessment using GPR: System configuration and defects' characterization. In *Construction Research Congress 2024*, 1210–1219.
- [3] Morris, Tyler, **Wang, Mengjun**, Li, Yan, Liu, Songyan, Li, Shuai, and Zhao, Xiaopeng. (2023). Awareness and acceptance of emerging technology and quadruped robots in dementia care: A survey study. *Proceedings of the AAAI Symposium Series*, 2(1), 187–191.
- [2] Hu, Da, **Wang, Mengjun**, and Li, Shuai. (2023). 3D object detection and localization within healthcare facilities. In *2023 Winter Simulation Conference (WSC)*, 2710–2721.
- [1] **Wang, Mengjun**, Hu, Da, Li, Shuai, and Cai, Jiannan. (2022). Urban subsurface mapping via deep learning-based GPR data inversion. In *2022 Winter Simulation Conference (WSC)*, 2440–2450.

Patents

- [1] Li, Shuai and **Wang, Mengjun**. (2026). Robotic AI Agent for Construction Guidance, Monitoring, and Documentation. PCT Patent Application No. PCT/US2026/028146, filed May 14, 2026.

Technical and Research Reports

- [2] Li, Shuai, Moslehy, Amirsalar, Hu, Da, **Wang, Mengjun**, Wierschem, Nicholas, Alshibli, Khalid, and Huang, Baoshan. (2022). Drones and Other Technologies to Assist in Disaster Relief Efforts. Tennessee Department of Transportation.
- [1] Sosoo, Kelvyn, Ouyang, David, and **Wang, Mengjun**. (2019 – equal contribution). License Plate Recognition and Matching Using Neural Networks. RECSEM REU Project Report, University of Tennessee.

RESEARCH LEADERSHIP & FUNDED PROJECT EXPERIENCE

- **Task-Grounded Physical AI and Spatial Co-Pilots for Human and Robotic Field Work** 2025 – Present
Research Lead; NSF National I-Corps Entrepreneurial Lead, University of Florida
 - Lead research connecting BIM-based task intent, workplace guidance, and task-relative robotic execution for field installation and assembly.
 - Served as Entrepreneurial Lead in an NSF National I-Corps cohort, conducting 100+ stakeholder interviews across construction and related project-based industries and leading cohort presentations and reporting.
 - This work contributed to a \$25,000 UF Innovate Innovation Fund award to the advisor-led team and a PCT patent application on robotic AI agents for construction guidance, monitoring, and documentation.
- **Multimodal Immersive Training to Develop Expertise in Spatial-Temporal Reasoning for Fire Investigations** 2026 – Present
Research Collaborator; National Science Foundation (NSF); UF PI: Shuai Li
 - Develop an immersive fire-investigation interface on an existing Unity-based fire simulation that lets learners label and link evidence in 3D and records their interactions for analysis of spatiotemporal reasoning.
- **FW-HTF-R: FAIR4WISE — Future AI and Robotics for Women in Smart Engineering** 09/2022 – 08/2025
Graduate Researcher; National Science Foundation (NSF); PI: Shuai Li
 - Contributed to robot teleoperation system design and implementation for human-centered construction robotics research.
- **CRII: CPS — Coupling Subsurface Features with Connected Autonomous Vehicles as Networked Cyber-Physical Systems for Reciprocal Mapping and Localization** 2021 – 2022
Graduate Researcher; National Science Foundation (NSF); PI: Shuai Li
 - Developed simulated GPR data and labels, deep-learning-based underground infrastructure detection, aboveground 3D reconstruction, and validation experiments for reciprocal mapping and localization.
 - Participated in a rapid-response GPR field survey at an East Tennessee mine-collapse site to collect subsurface data and assess potential additional voids near a residence.

- **Drones and Other Technologies to Assist in Disaster Relief Efforts** 2021 – 2022
Graduate Researcher; Tennessee Department of Transportation; PI: Shuai Li
 - Supported disaster-response research through scene exploration, literature review, and technical report development for drone-enabled sensing and mission planning.
- **RECSEM Research Experiences for Undergraduates (REU)** 05/2019 – 08/2019
Undergraduate Researcher; NSF-sponsored program, University of Tennessee; PI: Kwai Wong
 - Developed license-plate recognition and cross-camera matching workflows to infer vehicle speed and route information from highway camera data.

HONORS AND AWARDS

- **Third Place, ASCE Global Robotics and Automation Challenge** 2025
American Society of Civil Engineers (ASCE); individual/lead entry
- **Third Place, ESSIE Research Symposium** 2025
Engineering School of Sustainable Infrastructure & Environment, University of Florida; poster competition
- **2024 Best Paper Award, Journal of Computing in Civil Engineering** 2024
American Society of Civil Engineers (ASCE); coauthor of winning paper
- **Second Place, ASCE Data Challenge** 2024
American Society of Civil Engineers (ASCE)
- **Graduate Student Senate Travel Award** 2024
University of Tennessee
- **Graduate Top-Off Fellowship** 2023 – 2024
University of Tennessee; \$10,000/year
- **Outstanding Graduate of Hunan Province** 2020
Hunan Province, China
- **National Encouragement Scholarship** 2017, 2018, 2019
China
- **First Prize Scholarship** 2017, 2018, 2019
Changsha University of Science and Technology

TEACHING EXPERIENCE

- **Full-Time Primary School Teacher** 08/2020 – 08/2021
Grade 5 Mathematics, English, and Chinese; Zhongwan Primary School, China
 - Taught approximately 20 class periods per week in a rural primary-school setting.
 - Adapted instruction for a five-student class with different levels of preparation and progress.
- **Graduate Teaching Assistant** Spring 2022
CE 441/448: (Honors) Construction Engineering and Management II, University of Tennessee
 - Graded homework assignments and examinations.
- **Graduate Teaching Assistant** Fall 2021
CE 210: Geomatics, University of Tennessee
 - Graded homework assignments, laboratory reports, and examinations.

STUDENT MENTORING & STEM OUTREACH

- **Primary Research Mentor, Student Science Training Program (SSTP)** Summer 2026
University of Florida; faculty PI: Shuai Li
 - Mentored five rising 12th-grade students during an intensive seven-week, full-time summer research program focused on space construction and robotics.
 - Guided five individual research projects from problem definition and experimentation through posters, oral presentations, final reports, and abstract submissions to the 2026 Florida Space Summit.
- **Graduate Student Mentor, Senior Design Project** Fall 2024
University of Tennessee; faculty advisors: Jindong Tan and Shuai Li
 - Mentored a five-student senior design team developing a low-cost adaptive fin-ray gripper for industrial component pick-and-place.
 - Coordinated weekly meetings, reviewed progress, worked with faculty advisors on technical issues, and supported design iteration and problem solving.

- **Research Project Mentor, HITES¹² Pre-College Engineering Program** Summer 2024
University of Tennessee; faculty leads: Shuai Li and Jindong Tan
 - Co-mentored two four-student teams developing *AI Laser Projection and Construction Assistant (ALPCA)* and *Robotic Arm Manipulations Applied in Surgery (R.A.M.)*.
 - Guided students through hands-on implementation, experimental evaluation, research poster development, and presentation of their results.
- **Undergraduate Research Mentor** 2023 – 2024
University of Tennessee; construction automation and spatial augmented reality research
 - Mentored three undergraduate researchers on projects in projector calibration, chatbot interfaces, and laser-file processing, with individual research engagements ranging from six to eighteen months.
- **Robotics Outreach and Community Engagement** 2023
University of Tennessee
 - Participated in robotics demonstrations using Pepper and quadruped robots at UT Engineers Day, engineering conference events, and a Tennessee Alzheimer’s community event; collected and processed participant survey data for a study of awareness and acceptance of emerging technologies and quadruped robots in dementia care.

PEDAGOGICAL TRAINING & CERTIFICATIONS

- **CIRTL Associate Certificate** 2026
University of Florida Center for Teaching Excellence
- **Preparing Future Faculty** Fall 2025
University of Florida
- **Teacher Qualification Certificate, Junior Secondary School Mathematics** 2021
China

SELECTED RESEARCH PRESENTATIONS

- **Active Structured-Light Localization for Robotic Construction on Lunar Surface** 02/2026
ESSIE Research Symposium, University of Florida; poster presentation
- **Embodied Spatial Intelligence through AI-Enabled Robotic Laser Networks for Lunar Construction and Manufacturing** 10/2025
University of Florida Graduate School 3-Minute Thesis Competition
- **AI-Powered Laser Guidance: Revolutionizing Assembly in Modern Construction** 02/2025
ESSIE Research Symposium, University of Florida; poster presentation
- **Robotic Assembly of Interlocking Blocks for Construction Based on Large Language Models** 03/2024
Construction Research Congress 2024, Des Moines, IA; oral presentation
- **3D Object Detection and Localization within Healthcare Facilities** 12/2023
2023 Winter Simulation Conference, San Antonio, TX; oral presentation
- **Urban Subsurface Mapping via Deep Learning-Based GPR Data Inversion** 12/2022
2022 Winter Simulation Conference, Singapore; oral presentation
- **An Integrated Subsurface Mapping and Localization System** 09/2022
ISSE Annual Research Conference, University of Tennessee; poster presentation

ACADEMIC & PROFESSIONAL SERVICE

- **Journal Reviewer**
Automation in Construction; Journal of Computing in Civil Engineering;
Frontiers of Engineering Management; Engineering Applications of Artificial Intelligence
- **Conference Reviewer**
ASCE CRC 2024; ASCE I3CE 2024; ASCE I3CE 2025; 2026 Florida STEM Symposium
- **Senator** 03/2025 – 02/2026
University of Florida Student Government
- **Graduate Committee Member** 04/2025 – 03/2026
University of Florida AI2 Student Club Cabinet